ChatGPT Access and Writing Skill: A Bayesian Reanalysis of an Online Randomized Experiment

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This article is a preprint and has not been peer reviewed. It reports a secondary analysis of the original authors' archival data (Noy, 2023). I did not collect the data and have no affiliation with the original authors. This reanalysis was not registered. The analysis protocol, its dated record of deviations, and the analysis code are kept in a private repository and are available from me on reasonable request.

I have no conflicts of interest to disclose. This research received no specific funding. Under the CRediT taxonomy, I contributed Conceptualization, Methodology, Formal analysis, Investigation, Writing–original draft, and Visualization. During the preparation of this work I used large language models accessed through Cursor (Anthropic Claude, xAI Grok, OpenAI GPT, and Google Gemini) to draft and revise text and check reported statistics; a word processor with Clippy and autocorrect; and statistical software. I reviewed, edited, and verified the output and take full responsibility for the content. These tools are not authors.

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Abstract

Generative writing tools are entering professional work, and their effects differ across people. I reanalyze a published online randomized experiment in which college-educated professionals in six writing-intensive occupations completed two incentivized writing tasks, and a randomly assigned half were given access to ChatGPT before the second task (Noy & Zhang, 2023a). I ask whether access weakens the link between first-task and second-task writing grades, whether that weakening is stronger for writing than for content, whether access narrows differences between people more for originality than for writing quality, and how access relates to felt effectiveness and enjoyment on the task. Grades and ratings are standardized; the analysis plan, fixed beforehand, retests hypotheses already tested on these data. Bayesian regression models with weakly informative priors are fit to the linear and convex incentive conditions. Each estimate is a posterior median with a 95% equal-tailed credible interval and the posterior probability of the predicted direction, *P*(direction), with no decision cutoffs. Under access, second-task writing grades depended less on first-task grades, −0.159 [−0.317, −0.002], with *P*(direction) = .976; the data did not distinguish content from writing interactions, or originality from writing spread ratios. Access was associated with higher enjoyment (0.44 [0.29, 0.60]); felt effectiveness was 0.12 [−0.06, 0.29], *P*(direction) = .905. After a disclosed code correction, two-month workplace use of ChatGPT, the secondary outcome, was 13.4 [3.0, 23.5] percentage points higher under access. For organizations, results suggest one structured encounter was associated with higher enjoyment and followed by more use at work.

Keywords: generative artificial intelligence, online randomized experiment, Bayesian estimation, writing quality, technology adoption

ChatGPT Access and Writing Skill: A Bayesian Reanalysis of an Online Randomized Experiment

Professionals are already writing with a generative model in the loop. For organizational and occupational psychologists, two questions follow. The first concerns the work: when a person is handed the tool, does the product get better? The second concerns the worker: after one structured encounter with the tool, does the person keep using it on the job? The first question is about performance and the second about behavior change. Both are usually studied with observational data, in which people who choose a tool differ from people who do not. Random assignment rules out that self-selection.

One such experiment assigned access to ChatGPT at random (Noy & Zhang, 2023a). College-educated professionals recruited on Prolific, working in marketing, grant writing, consulting, data analysis, human resources, and management, completed two occupation-specific writing tasks under a monetary incentive. After the first task, a random half were instructed to sign up for ChatGPT and permitted to use it on the second task; the other half signed up for a document editor. In the main incentive conditions, the access group spent less time and received higher grades (Noy & Zhang, 2023a, p. 188). A follow-up survey two months later invited everyone who completed the main survey (Noy & Zhang, 2023b, p. 6), and 42% of the access group and 27% of the comparison group reported using ChatGPT in their jobs in the past week (Noy & Zhang, 2023a, p. 190). People whose first essay received a lower grade gained more on the second (Noy & Zhang, 2023a, p. 188; Figure 2, p. 190). The data and code are public (Noy, 2023).

Studies of generative tools in other kinds of work report average gains that are uneven across workers. Among 5,172 customer-support agents, access to a conversational assistant

raised issues resolved per hour by 15% on average, and the gains accrued disproportionately to less experienced and lower-skill agents (Brynjolfsson et al., 2025, pp. 889–890). In an experiment with 758 consultants, those given access to an artificial intelligence (AI) tool completed 12.2% more tasks, 25.1% more quickly, with higher quality on tasks inside the tool's capabilities, but were 19% less likely to reach a correct solution on a task chosen to lie outside them (Dell'Acqua et al., 2026, p. 403), and relatively lower-skilled consultants gained the most (Dell'Acqua et al., 2026, pp. 410–411). Developers with an AI pair programmer completed a coding task 55.8% faster, and those with less programming experience benefited the most (Peng et al., 2023, pp. 1–2). Writers given story ideas from a large language model, especially less creative writers, produced stories rated more creative, better written, and more enjoyable, but the stories were more similar to one another (Doshi & Hauser, 2024, p. 1).

Evidence from learning and writing settings points to costs that average gains hide. High school students with GPT-4 access improved their grades while they had it, but when access was taken away, students who had used an interface that mimicked standard ChatGPT performed 17% worse than students who never had access (Bastani et al., 2025, p. 1). University students who wrote with ChatGPT improved their essay scores more than comparison groups, without a difference in knowledge gain, knowledge transfer, or intrinsic motivation (Fan et al., 2025). In a study of 54 participants writing essays, reported in a preprint that has not been peer reviewed, those who used a language model showed the weakest neural connectivity, reported low ownership of their essays, and were less able to quote from essays they had just written (Kosmyna et al., 2025, p. 2). These studies show that an average effect of access hides gains and costs concentrated among some people, yet most compare subgroups defined one at a time, such as experience or skill, and fewer ask whether access changes how far people differ from one

another (Doshi & Hauser, 2024, p. 1; Kosmyna et al., 2025, p. 2; Noy & Zhang, 2023a, pp. 187–189).

This study asks about the pattern of effects across people. Access to the same tool may lift some writers a great deal and leave others unchanged; whether it also makes the work of different writers more alike is a possibility this study tests. For organizations deciding whom to train, whom to give access first, and whom to support, the pattern of effects matters more than the average. I therefore ask how access changes the relation between a person's earlier and later writing, how it changes the spread of quality across people, and how it relates to how effective people felt and how much they enjoyed the task. Whether people were using ChatGPT or a similar chatbot at work in the past week, at the two-month follow-up, is a secondary question. These questions are the five hypotheses fixed in the analysis plan, four primary and a fifth, on enjoyment, that the plan labels secondary. Aims taken up and withdrawn in earlier revisions are listed in the Appendix (Earlier Aims Withdrawn and the Hypotheses Restored).

Theoretical Background

Who Gains From a New Resource

Conservation of resources theory holds that people seek to obtain, protect, and invest resources, and that resource gains shape later performance and well-being (Hobfoll, 1989; Hobfoll et al., 2018). The theory supports two opposite predictions about who gains most from a new resource. On one reading, a tool that supplies a competent draft substitutes for resources a person lacks, so people with weaker baseline writing, less experience with the task, or less time should gain most (compensation). On another, resources travel in caravans: people already rich in skill and experience are better placed to direct, judge, and edit the tool's output, so they should gain most (accumulation; Hobfoll et al., 2018). The original finding that lower first-essay grades

went with larger gains (Noy & Zhang, 2023a, p. 188) favors compensation, and the hypotheses below take that direction; a result in the opposite direction would favor accumulation.

Work-design research adds that the effect of a digital technology depends on how it is inserted into the task and on the person's skill, autonomy, and feedback (Parker & Grote, 2022). In this experiment, people were free to use the tool however they liked on one task, so differences between people in the effect reflect access as offered, not any prescribed way of using the tool.

Hypothesis 1: The association between first-task and second-task writing grades will be weaker in the access group than in the comparison group (a negative interaction between condition and the centered first-task writing grade).

If domain judgment is the part of the work the tool does not replace, the baseline gradient for content should be less negative than for writing, so that high-baseline writers keep their edge on substance and low-baseline writers gain most on prose. The original supplement found that ChatGPT "does not have greater benefits for treated respondents whose writing skills are poor relative to their other skills" (Noy & Zhang, 2023b, Figure S.4(b), p. 91), so this hypothesis rests on that reasoning and tests it directly.

Hypothesis 2: The same interaction will be algebraically larger (less negative, or positive) for content grades than for writing grades.

If access pulls essays toward a shared style, unusual choices are smoothed away, so the spread of originality should fall more than the spread of writing quality, which tightens only because weak prose is repaired.

Hypothesis 3: Access will reduce the spread of second-task originality grades across people more than it reduces the spread of writing grades (the access-to-comparison log ratio of standard deviations will be lower for originality than for writing).

Who Keeps Using It

Whether a person keeps using a tool after a first encounter is a motivational question. Self-determination theory distinguishes behavior experienced as volitional and competence-enhancing from behavior controlled from outside (Gagné & Deci, 2005; Ryan & Deci, 2000). A structured first use could raise later adoption most for people whose first encounter is a competence-relevant success, and least for people who value ownership of their writing or who do not need the help. Enthusiasm, contentment, and meaningfulness at work are separable, so liking a task and finding it meaningful are different outcomes (Monnot & Beehr, 2014), and the meaning of work has its own sources and mechanisms (Rosso et al., 2010, pp. 91–93). These ideas motivate treating felt effectiveness and enjoyment on the task as outcomes of a first encounter and later use as a secondary outcome.

Under compensation, the tool supplies most to writers whose first essays were graded lower, so their felt effectiveness and enjoyment on the second task should rise most.

Hypothesis 4: Access will be associated with higher felt effectiveness on the second task, conditional on felt effectiveness on the first, and that difference will be larger for people with lower first-essay overall grades.

Hypothesis 5: Access will be associated with higher enjoyment of the second task, conditional on enjoyment of the first, and that difference will be larger for people with lower first-essay overall grades. Hypothesis 5 is a secondary hypothesis.

The secondary outcome is the effect of access on the proportion of people who reported using ChatGPT or a similar chatbot in their job in the past week at the two-month follow-up.

The Present Study

The questions are stated as directional hypotheses. Each names an estimand and its predicted direction, and each is reported with its full posterior rather than judged against a cutoff. Differences by condition are intention-to-treat effects; interactions with first-task grades describe who benefits and are not causal effects of those grades.

Method

Transparency and Openness

I report how the analysis sample was determined, all data exclusions, all manipulations, and all measures, following the Journal Article Reporting Standards for quantitative research (Appelbaum et al., 2018) and the Consolidated Standards of Reporting Trials (CONSORT) 2025 guideline for randomized trials (Hopewell et al., 2025). The original experiment was registered by its authors in the American Economic Association registry for randomized controlled trials (AEARCTR-0010882; Noy & Zhang, 2023a). The data are the original authors' archival files on the Open Science Framework (Noy, 2023). This reanalysis was not registered, and the word registration in this article refers only to the original authors' entry. The analysis protocol for this reanalysis was fixed and time-stamped before the planned real-data analysis, and the protocol, code, and dated deviation record are kept in a private repository and are available on request. The original authors obtained consent within the survey (Noy, 2023). This secondary analysis does not redistribute participant-level data. The original experiment was approved by the MIT Committee on the Use of Humans as Experimental Subjects (protocol no. 2212000849; Noy & Zhang, 2023a, Acknowledgments). This reanalysis used only those publicly posted archival files,

in which the depositors report replacing Prolific IDs with hashes (https://doi.org/10.17605/OSF.IO/XD7QW); no participants were contacted, and this secondary analysis was not submitted as a new human-subjects study. Analyses used R (Version 4.5.2) with brms (Version 2.23.0; Bürkner, 2017) on the Stan backend rstan (Version 2.32.7; Carpenter et al., 2017), posterior (Version 1.7.0), and mice (Version 3.19.0; van Buuren & Groothuis-Oudshoorn, 2011), and Python (Version 3.14.2) with pandas (Version 3.0.0), NumPy (Version 2.4.1), and DuckDB (Version 1.5.6) to build the analysis files from the archive. Because this study is a secondary analysis of existing, publicly posted data (Noy, 2023), neither participants nor members of the public were involved in its design.

Participants

Participants were college-educated professionals recruited on Prolific in six occupations: marketing, grant writing, consulting, data analysis, human resources, and management (Noy & Zhang, 2023a). The original authors screened participants on self-reported occupation and preferentially recruited those with college degrees and full-time employment; after the first 20 data-analyst responses, data analysts were screened in only if they reported writing code to analyze data as one of their main job tasks, and respondents who twice failed the comprehension checks on the instructions were screened out (Noy & Zhang, 2023b, pp. 2, 8). Neither the article nor the supplement reports a screen on country of residence, and the archival survey files contain no country item (Noy, 2023), so the participants' countries cannot be described. The original authors' published analysis sample is 453 participants (Noy & Zhang, 2023a, p. 187), whereas the supplement reports that recruitment stopped at 450 responses (Noy & Zhang, 2023b, p. 8); neither source explains the difference. I reconstruct the published analysis sample from the archival files using the original authors' exclusion rules (unfinished responses, nonconsent,

duplicate entries, named exclusions, a quality check, and participants who also served as graders before writing), and the participant flow diagram (Figure 1) shows, by condition, the randomized participants who remain, the primary sample and the exact-time condition, and the number in the primary sample with each outcome available. The primary sample is the participants in the linear and convex incentive conditions, 359; the exact-time condition is described but not modeled (see Appendix, Exploratory Analyses and Model Checks Not Built). Each complete-case model analyzes the participants in the primary sample with the outcome and all baseline covariates observed: n = 352 for Hypothesis 1, 352 for Hypothesis 2, 353 for Hypothesis 3, 352 for Hypothesis 4, 352 for Hypothesis 5, and 276 for adoption; the adoption count is for the complete-case adoption model, and the primary adoption estimate, by multiple imputation, uses all 359 participants in the primary sample. Participants without the two-month follow-up are not scored on adoption and are described as an attrition subset.

Age, gender, and race or ethnicity are not in the released survey file, so they cannot be reported. Table 1 reports, by condition, the characteristics recorded before assignment: occupation, employment status, tenure in the occupation, weekly hours, prior experience with a similar task, rated realism of the task, felt effectiveness and enjoyment on the first task, task order, first-essay writing, content, originality, and overall grades, and prior use and awareness of ChatGPT. For description only, it also shows reported ChatGPT use on the second task and completion of the two-month follow-up, both recorded after assignment.

Because the sample is fixed by the archive, no prospective sample-size calculation was possible. Precision is reported directly as the width of each interval.

Design and Randomization

The design is a two-group, parallel online randomized experiment with a pretest (first task) and posttest (second task) in a single online session, with a follow-up survey about two months later. Access condition was assigned after the first task and its follow-up questions (Noy & Zhang, 2023b, Figure S.1, p. 5). Payment scheme was randomized at the start of the survey: a linear bonus per grade point, a convex bonus with an extra payment for high grades, or an exact-time condition that fixed time on each task (Noy & Zhang, 2023b, p. 3). For payment scheme, the first 189 respondents were assigned with a Qualtrics random-integer function; because it did not produce the intended 40/40/20 split, the original authors then used the Qualtrics randomly present element feature with allocations of 36% linear, 42% convex, and 22% exact-time (Noy & Zhang, 2023b, p. 8). Task order (which of two occupation-specific prompts came first) was also randomized (Noy & Zhang, 2023b, Figure S.1, p. 5). Half of the participants were assigned to the access condition and half to the comparison condition (Noy & Zhang, 2023a, p. 187). Neither the article nor the supplement describes how the assignment to condition was generated, and the archival files record each participant's condition but no random draw for it (Noy, 2023). The randomization procedure, allocation concealment, and implementation were the original authors'; I had no role in them. Participants could not be blinded to condition. Graders were a separate Prolific sample who rated essays without being told the writer's condition.

Intervention

After the first task, participants in the access condition were instructed to create a ChatGPT account and told they could use it on the second task. Participants in the comparison condition were instructed to create an account on Overleaf, a document editor, and were not given ChatGPT access for the second task (Noy & Zhang, 2023a). Use of ChatGPT was

permitted, not required, and how participants used it was not controlled. The estimand is therefore the effect of being offered access (an intention-to-treat effect), not the effect of using the tool. Self-reported use on the second task is described but does not define the groups, because use was not randomized.

Measures

Writing Quality

Graders who were not told the writer's condition rated each essay on a 1–7 scale for overall quality, writing quality, content quality, and originality. Following the original authors' code (Noy, 2023), every grader whose grades correlate below .10 with those of the other graders who saw the same essays was screened out (in the original report, 23 of 148 graders; Noy & Zhang, 2023b, p. 7), and each person-level score was the mean of the remaining graders' ratings. The writing, content, and originality grades of the second essay are the outcomes for Hypotheses 1–3, and the same grades of the first essay are baseline covariates. In the original report, each essay had three graders, and the mean within-essay correlation among graders was .44 (Noy & Zhang, 2023a, p. 187). Grading rows marked as retries were dropped before the mean was taken. Interrater agreement on the analysis sample is reported in the Table 1 note as intraclass correlation coefficients, ICC(1) and ICC(k).

Workplace Adoption

At the two-month follow-up, participants reported whether they had used ChatGPT or a similar chatbot in their job in the past week. A *yes* response is coded 1, a *no* response 0, and any other response is missing. The original authors launched the follow-up at a referee's suggestion, to test whether effects on beliefs about automation persisted, and took the opportunity to ask

again about ChatGPT use at work (Noy, 2023, ReadMe for Replication Pkg.pdf, p. 8). Because the item was added after the original design, it is a secondary outcome here.

Baseline Covariates

The baseline covariates, all recorded before assignment, are occupation, task order, payment condition, the first-essay writing, content, originality, and overall grades, and felt effectiveness and enjoyment on the first task. Employment status, tenure, weekly hours, prior experience with a similar task, and rated task realism are described in Table 1. The task timers (the Qualtrics page timers and the self-reported minutes on each task) are excluded from every model; the second-task timers were recorded after assignment and could transmit part of the effect of access. A posttask item asking whether ChatGPT was used on the second task (Noy, 2023, ReadMe for Replication Pkg.pdf, p. 4) is reported by condition for description only and is not a covariate (see Appendix, Posttask ChatGPT-Use Items Excluded as Covariates). Before the first task, participants were asked which of six software tools they were aware of and which they had used (Google Drive, Tableau, ChatGPT, Overleaf, Jasper, and Grammarly; Noy, 2023, ReadMe for Replication Pkg.pdf, p. 3). Two indicators were built from these items: prior use of ChatGPT and awareness of ChatGPT. Each is coded 1 if ChatGPT was selected, 0 if the item was answered without selecting ChatGPT, and missing if the item was not answered; missing values are imputed within the adoption model's multiple imputation, following the analysis plan. Both are prespecified baseline covariates in the adoption model only (Table 1).

Felt Effectiveness and Enjoyment

Two single items, each rated from 1 to 10, ask how skilled or effective the person felt on the task just completed and how much the person enjoyed it. They are stored in the archival data set under the names *self-efficacy* and *job satisfaction*, but they are task-specific ratings, not scales

of general self-efficacy or job satisfaction, and they are described here by their content. Ratings of the first task are covariates; ratings of the second task are the outcomes for Hypotheses 4 and 5.

Procedure

In one online session, participants consented, completed the first writing task, rated how much they enjoyed it and how effective they felt, were assigned to condition and set up the assigned tool, completed the second task, and rated it again. A separate pool of graders rated the essays. About two months later, participants were invited to a follow-up survey that asked about use of ChatGPT in their jobs (Noy & Zhang, 2023a). Main data collection ran from January 27 to February 21, 2023 (Noy & Zhang, 2023a, p. 187), and the two-month follow-up was open from April 19 to April 29 (Noy & Zhang, 2023b, p. 6). Recruitment ended when the original authors ran out of funds at 450 responses, short of the 600 they had registered, because screening and grading cost more than expected (Noy & Zhang, 2023b, p. 8). The original authors give a budget reason for the stop and report no interim analysis or statistical stopping rule (Noy & Zhang, 2023b, p. 8), and this reanalysis has no interim analysis: the outcomes were analyzed in a planned real-data analysis on October 30, 2026, and, for the adoption analyses, in a corrected follow-up analysis the same day after a disclosed postfreeze correction (see Appendix, Adoption Analyses Reestimated After a Code Correction). No harms or adverse-event data were collected; neither the article nor the supplement reports harms collection, the supplement’s protocol deviations (§A.3) include no harms-related procedure (Noy & Zhang, 2023a, 2023b), and the replication package has no harms variable (Noy, 2023).

Analysis Plan

This plan was written before any model under it was fit to the real outcomes. Models were first run on synthetic data with the same structure. On October 29, 2026, I ratified version 3 of the analysis plan and authorized a planned real-data analysis; the plan that was frozen and run is version 3d, dated October 30, 2026, which incorporates refinements made by AI agents working under my direction, and the AI agents applied my October 29 authorization to it without a separate ratification of version 3d (see Appendix, Plan Version Ratified and Plan Version Run). The real data were analyzed in a planned real-data analysis on October 30, 2026, and, for the adoption analyses, in a corrected follow-up analysis the same day after a disclosed postfreeze correction that I authorized for that read (see Appendix, Adoption Analyses Reestimated After a Code Correction).

Scale of Outcomes and Coefficients

First-task writing, content, and overall grades were centered at the primary-sample mean and divided by the comparison group's standard deviation of the same grade, second-task writing and content grades were rescaled with the mean and standard deviation of the matching first-task grade, and second-task originality grades and the felt effectiveness and enjoyment ratings of both tasks were standardized to a mean of 0 and a standard deviation of 1 in the primary sample. As context, and not as a decision rule, standardized differences between conditions can be compared with the benchmarks proposed for causal studies of education interventions on achievement, in which effects below 0.05 standard deviations are small, effects from 0.05 to below 0.20 are medium, and effects of 0.20 or more are large (Kraft, 2020).

Models

All models were Bayesian regressions fit in brms (Bürkner, 2017) on the primary sample. All except the Hypothesis 3 location-scale model include occupation, task order, and payment condition as covariates. For Hypothesis 1, the second-task writing grade was regressed on condition, the centered first-task writing grade, and their product. For Hypothesis 2, writing and content grades were modeled jointly with a residual correlation; the estimand is the content interaction minus the writing interaction. Hypothesis 2 is theory-driven under conservation-of-resources compensation; the original frequentist supplement finding (Noy & Zhang, 2023b, Figure S.4(b), p. 91) is not used as a prior for this Bayesian estimand. For Hypothesis 3, a location-scale model let the mean and the standard deviation of each grade differ by condition, with no other predictors, so that each standard deviation was the spread of grades within a condition, as the hypothesis defines it; the estimand is the originality log ratio minus the writing log ratio. For Hypotheses 4 and 5, second-task felt effectiveness and enjoyment were regressed on condition, the first-task rating, the centered first-essay overall grade, and the product of condition and that grade. Because grades lie between 1 and 7 and ratings between 1 and 10, ordinal and censored models were fit as exploratory analyses. For adoption, a logistic model of two-month use on condition, prior use of ChatGPT, awareness of ChatGPT, task order, payment condition, and occupation yielded the risk difference by averaging predictions over the sample.

Priors

The coefficient for condition receives only a weakly informative prior, so it is not shrunk toward zero: Normal(0, 1) on the standardized scale, or Normal(0, 1.5) on the log-odds scale for adoption, following the weakly informative default priors for regression models (Gelman et al., 2008). The condition-by-baseline interactions receive Normal(0, 0.5) priors, which place about

95% of their mass between −1 and 1; the coefficients of the covariates receive Normal(0, 1) priors in every model, including the adoption model; and the residual standard deviations in the models for Hypotheses 1, 2, 4, and 5 receive Exponential(1) priors. Residual correlations receive an LKJ(2) prior. In the Hypothesis 3 location-scale model, the condition coefficient receives a Normal(0, 0.5) prior on the log-*SD* scale. This Hypothesis 3 log-*SD* prior places about 95% of its mass on multiplicative factors for the access-to-comparison ratio of standard deviations between about 0.38 and 2.66. The exploratory models use the priors listed in Table S2 of the Online Supplement; they differ from those above in that the ordinal models for Hypotheses 4 and 5 give the condition coefficient a Normal(0, 1.5) prior and the condition-by-grade interaction a Normal(0, 0.75) prior, and the grader-level model gives the standard deviations of its grader and person random intercepts Exponential(2) priors. The intercepts, including the standard deviation intercepts, keep the brms defaults, which brms sets from the data; the priors used are recorded with each fitted model. Prior sensitivity was assessed (Depaoli & van de Schoot, 2017) by doubling and halving the prior scales for the condition coefficient, the condition-by-baseline interaction, and the Hypothesis 3 log-*SD* condition coefficient. A further check replaced the priors on the condition coefficients, the condition-by-baseline interactions, and the Hypothesis 3 log-*SD* condition coefficient with diffuse Normal(0, 10) priors.

Estimation and Checks

Each model was run with 8 chains of 4,000 warmup and 4,000 sampling iterations, a target acceptance rate of .99, and a maximum tree depth of 15. A model was reported only if every parameter had *R*-hat of 1.01 or less (Vehtari et al., 2021), bulk and tail effective sample sizes of at least 400, no divergent transitions, an energy Bayesian fraction of missing information (E-BFMI) of at least 0.3, and a Monte Carlo standard error (MCSE) of the posterior median

below 1% of the posterior standard deviation for each reported estimand. The MCSEs of the 2.5% and 97.5% interval endpoints and of the posterior probability of direction are computed but are not used as a criterion. Otherwise it was rerun once with doubled warmup and sampling iterations and a target acceptance rate of .999, and if it still failed, the estimand was reported as not estimable with its diagnostics. For every fit, the sampler settings, number of attempts, largest *R*-hat, smallest bulk and tail effective sample sizes, divergences, smallest E-BFMI, and the ratio of the MCSE of the posterior median to the posterior standard deviation were reported in Table S1 of the Online Supplement, the priors as coded in Table S2, and each attempt of the planned real-data analysis in Table S3. Ordinary least squares estimates were reported beside the posteriors as a robustness check, with HC3 heteroskedasticity-consistent standard errors for Hypotheses 1, 4, and 5 and a person bootstrap for all five hypotheses.

Reporting

Every estimand was summarized by its posterior median, its 95% equal-tailed credible interval, and the posterior probability that it lies in the predicted direction. Estimates for Hypotheses 1, 2, 4, and 5 were on the standardized scale defined above, the Hypothesis 3 estimate was a difference of log standard-deviation ratios, and adoption was a risk difference in percentage points; ordinary least squares estimates were reported beside them, with HC3 intervals for Hypotheses 1, 4, and 5 and person-bootstrap intervals for all seven estimands. No adjustment for multiplicity was made across the five hypotheses: the posterior probabilities of direction were unadjusted and descriptive, and a high probability of direction indicates the sign of an effect, not that an effect of practical size exists. The plan set no smallest effect size of interest. No decision cutoff was used, and results were described by their size and precision. Following the Bayesian Analysis Reporting Guidelines (Kruschke, 2021), the software, priors,

and sensitivity results were reported in the text, and the priors as coded and the convergence diagnostics of every fit in Tables S1 to S3 of the Online Supplement. All counts were reported exactly (see Appendix, Exact Counts for Public Data).

Missing Data

Models for the grade and rating outcomes (Hypotheses 1–5) were fit to complete cases on the outcome, with all baseline covariates in the model (Little & Rubin, 2019). A participant missing a first-task grade or another baseline covariate was excluded from that model, and the number excluded was reported for each model. The baseline covariates were recorded before assignment, and complete-case analysis is unbiased when covariate missingness is independent of the outcome given the covariates (White & Carlin, 2010). For adoption, missing follow-up responses were multiply imputed (50 imputations), and the result was compared with complete-case estimates, a tipping-point analysis that shifted the imputed values, and bounds that trimmed the group with more respondents (Lee, 2009). Follow-up response was reported by condition (Table 2); no model compared it between conditions.

Exploratory Analyses

The exploratory analyses were the censored models for Hypotheses 1 and 3, the ordinal models for Hypotheses 4 and 5 (see Models), and a multilevel refit of the Hypothesis 1 model on the individual writing ratings, with grader and person random intercepts, fit to the ratings of both essays without restriction to the second essay (1,918 ratings of 705 essays), which was reported as an exploratory grader-level model and not as a check on Hypothesis 1. The earlier plan also listed the main models in the exact-time condition, the grade models refit with the grader screen turned off, the models within each task-order stratum, and a median-absolute-deviation

comparison for Hypothesis 3; these were not built under the frozen plan and were not reported (see Appendix, Exploratory Analyses and Model Checks Not Built).

Results

Participant flow and baseline characteristics appear in Figure 1 and Tables 1 and 2.

Hypotheses 1–5 are summarized in Table 3 and Figure 2. Each estimate below is a posterior median followed by its 95% equal-tailed credible interval in brackets, on the scales defined in Method. Every fit for Hypotheses 1–5, under the primary and the three alternative priors, met the convergence and Monte Carlo error criteria on its first attempt, with no divergent transitions and a largest *R*-hat of 1.001. In the access group, second-task writing grades depended less on first-task writing grades than in the comparison group: the condition-by-baseline interaction was −0.159 [−0.317, −0.002], with *P*(direction) = .976 (Hypothesis 1). The content interaction minus the writing interaction was −0.01 [−0.14, 0.11], with *P*(direction) = .423, an interval that includes zero, so the data do not distinguish the two interactions (Hypothesis 2). The originality-minus-writing difference in log standard-deviation ratios was −0.01 [−0.14, 0.12], with *P*(direction) = .576 (Hypothesis 3). The condition effect on felt effectiveness on the second task was 0.12 [−0.06, 0.29], with *P*(direction) = .905, an interval that includes zero, and the condition-by-grade interaction for felt effectiveness was −0.11 [−0.27, 0.06], with *P*(direction) = .894 (Hypothesis 4). Access was associated with higher enjoyment of the second task, 0.44 [0.29, 0.60], with *P*(direction) = 1.000, and the condition-by-grade interaction for enjoyment was 0.04 [−0.11, 0.19], with *P*(direction) = .299 (Hypothesis 5).

The sensitivity analyses specified in the analysis plan were the three alternative prior scenarios, the ordinary least squares check, and, for adoption, the complete-case, tipping-point, and trimming comparisons, which are reported with adoption below. Doubling the prior scales,

halving them, or replacing them with diffuse Normal(0, 10) priors changed the posterior medians for Hypotheses 1–5 little: the largest change was for Hypothesis 1 under halved scales, −0.147 [−0.301, 0.004], with *P*(direction) = .972, an interval whose upper bound lies just above zero and is at zero to two decimals, and every other median changed by less. Under the three alternative priors, the complete-case adoption risk difference from the corrected follow-up analysis ranged from 12.2 to 14.0 percentage points. The ordinary least squares estimates had the same signs as the posterior medians; for Hypothesis 1 the estimate was −0.17, with an HC3 95% confidence interval of [−0.35, 0.02] and a person-bootstrap interval of [−0.35, 0.02], and for the condition effect on enjoyment it was 0.45, with intervals of [0.28, 0.61] and [0.28, 0.61]. Among the exploratory models, censored models for the bounded grades gave a Hypothesis 1 interaction of −0.18 [−0.34, −0.01], with *P*(direction) = .983, and a Hypothesis 3 contrast of −0.03 [−0.24, 0.19], with *P*(direction) = .595. Ordinal models gave, on the log-odds scale, condition effects of 0.38 [0.00, 0.75], with *P*(direction) = .977, on felt effectiveness and 1.17 [0.79, 1.57], with *P*(direction) = 1.000, on enjoyment, and condition-by-grade interactions of −0.28 [−0.63, 0.07], with *P*(direction) = .937, and 0.14 [−0.22, 0.50], with *P*(direction) = .227, respectively. The grader-level model with grader and person random intercepts gave −0.09 [−0.18, −0.01], with *P*(direction) = .981, but because it pooled the ratings of the first and second essays, it is not a direct check on Hypothesis 1.

Two-month adoption appears in Table 4. These estimates come from the corrected follow-up analysis after the disclosed correction to the code that averages model predictions (see Appendix, Adoption Analyses Reestimated After a Code Correction). The adoption item was answered by 139 of 172 participants in the access group and 150 of 187 in the comparison group (Table 2). Over all 359 participants in the primary sample, the multiple-imputation risk

difference was 13.4 [3.0, 23.5] percentage points, with *P*(difference > 0) = .994, and model-standardized adoption was 42.2% in the access group and 28.8% in the comparison group. Among the 276 participants with the adoption item and every covariate observed, the complete-case risk difference was 13.5 [3.1, 23.7] percentage points, with *P*(difference > 0) = .995. The tipping-point analysis gave risk differences of 10.1 to 17.2 percentage points, and the trimming bounds, which assume that access moved follow-up response in one direction only, were 14.3 to 15.7 percentage points. The 50 imputations were drawn by chained equations, run separately within each condition, from a model with the adoption item (imputed by logistic regression), prior use and awareness of ChatGPT, task order, payment condition, occupation, the first-essay writing and overall grades, and felt effectiveness and enjoyment on the first task.

Discussion

Findings and Interpretation

This reanalysis asked whether access to ChatGPT changes the pattern of writing outcomes across people, beyond its average effect. Across the five hypotheses, the evidence was strongest for the writing-grade interaction (Hypothesis 1) and the enjoyment effect (Hypothesis 5), weaker for the felt-effectiveness effect (Hypothesis 4), and the estimates were near zero for the two contrasts between grade types (Hypotheses 2 and 3). Second-task writing grades depended less on first-task writing grades in the access group than in the comparison group: the condition-by-baseline interaction was −0.159 [−0.317, −0.002], with *P*(direction) = .976, so for each standard deviation higher on the first-task writing grade, the difference between conditions in the second-task writing grade was smaller. The upper bound of the interval lies just below zero and is at zero to two decimals, and the size of the estimate changed little under the alternative priors, the censored model, and ordinary least squares, although the least squares

intervals and the interval under halved prior scales included zero. The Hypothesis 2 contrast, a difference between two interactions on correlated grades, was expected to be imprecise; its estimate was −0.01 [−0.14, 0.11], with *P*(direction) = .423, so the data do not distinguish the content and writing interactions. The originality-minus-writing difference in the log ratio of spread in the access group to spread in the comparison group was −0.01 [−0.14, 0.12], with *P*(direction) = .576, so the data do not distinguish originality from writing grades on this ratio. Access was associated with higher enjoyment of the second task, 0.44 [0.29, 0.60], with *P*(direction) = 1.000, and the condition effect on felt effectiveness was 0.12 [−0.06, 0.29], with *P*(direction) = .905.

In relation to compensation and accumulation (Hobfoll et al., 2018), the writing interaction is in the direction compensation predicts, which is the direction the hypotheses took: the difference between conditions was larger for writers whose first essays were graded lower, as the original authors reported (Noy & Zhang, 2023a, p. 188). The felt-effectiveness interaction was −0.11 [−0.27, 0.06], with *P*(direction) = .894, and the enjoyment interaction was 0.04 [−0.11, 0.19], with *P*(direction) = .299, so the compensation pattern was clear for writing grades but not for either rating. Felt effectiveness, enjoyment, and adoption bear on whether a first encounter with the tool is one that people want to repeat (Ryan & Deci, 2000). I read the two ratings as reported reactions to the task rather than as measures of competence satisfaction, and no model estimated the difference between the condition effects on the two ratings. Two months later, the risk difference in use at work was 13.4 [3.0, 23.5] percentage points, with *P*(difference > 0) = .994, estimated in the corrected follow-up analysis after a disclosed code correction (see Appendix, Adoption Analyses Reestimated After a Code Correction). No model linked the task

reactions to later use; what the results show is that a single structured encounter was followed by higher enjoyment of the task and, two months later, by more use at work.

For organizations, the pattern of effects carries two practical points. The gain from access was largest for writers whose first essays were graded lower, so the case for early access and training is strongest for them rather than for the strongest writers, for whom the difference between conditions was smallest. One structured encounter was followed by higher enjoyment of the task and, two months later, by more use at work, so a single supported first use is a reasonable way to introduce the tool. Studies that measure ownership and meaningfulness, follow use beyond two months, and vary how the tool is inserted into the task (Parker & Grote, 2022) can test the mechanisms this design could not. In this experiment, access to ChatGPT weakened the link between earlier and later writing grades, was associated with higher enjoyment of the task, and was followed two months later by more use at work.

Limitations

I did not design or run the experiment, and earlier analyses within this project looked at overlapping outcomes: the hypotheses, as worded in the frozen plan, had been tested once on these outcomes in the October 22 analysis, before the plan was written, so the planned real-data analysis is a retest (see Appendix, Earlier Aims Withdrawn and the Hypotheses Restored). Demographics were not released. Each grade is the mean of a few graders, 2.72 per essay on average, on a bounded 1–7 scale, and agreement among graders was limited, ICC(k) = .63 for writing and .53 for originality, so compression near the ends of the scale and grader noise can shape slopes and spreads mechanically; the Hypothesis 1 and 3 contrasts therefore describe the graded scores, although few second-task writing grades lay at a bound (7 grades at the floor of 1 and 3 at the ceiling of 7, among 353) and the censored model gave a Hypothesis 1 estimate of the

same sign (see Results). Felt effectiveness, enjoyment, and adoption are single self-report items from participants who knew their condition, so the ratings are open to demand characteristics and are reported reactions to the task, not measures of competence satisfaction. Ownership and meaningfulness were not measured. Adoption is missing for participants who did not answer the follow-up. Conditioning on postassignment process variables would change the estimand (Montgomery et al., 2018), so those items stay out of the five planned models. Immediate task outcomes cannot establish durable workplace value; two-month adoption is observed only among follow-up responders (Lee, 2009; Noy & Zhang, 2023a). Missing baseline covariates are handled by complete cases in the grade and rating models, and adoption is compared across imputed and complete-case estimates (Little & Rubin, 2019; White & Carlin, 2010). Claims are limited to early-2023 ChatGPT among the archival Prolific occupations and may not generalize to current tools or workplaces.

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Appendix

Deviations From the Frozen Analysis Plan

Every departure from the first fixed plan for this reanalysis (October 22, 2026) is listed below, with its date where one is recorded.

Earlier Aims Withdrawn and the Hypotheses Restored

The October 22 plan had five hypotheses about first-essay grades, originality dispersion, and felt effectiveness and enjoyment. On October 24, the aims were briefly changed to heterogeneity and prediction: predicting adoption, predicting writing quality, and estimating with causal forests how the effect of access varies across people. The current plan returns to the five hypotheses, now estimated with Bayesian models. The October 22 analysis used the complete

archival data: the primary sample of 359 of the 453 published participants, built under that plan, with all outcomes, so the outcomes had been seen before the current plan was written. That analysis gave intervals including zero for Hypotheses 1–4 and a positive estimate for Hypothesis 5. Its outputs were withdrawn on October 24 and are not used here. The current plan and code were fixed after that run, so this is a reanalysis made after a first result was known. The current plan text is a reconstruction of a lost revision, reconciled to the locked analysis code.

Change in Time Spent Dropped

The October 22 plan included the change in minutes spent on the task as a replication outcome. It is dropped because it is built from the task timers, which are not used anywhere in the current plan, and the current plan contains no replication check, so the change in overall grade, described as a replication check in earlier drafts of this article, is not computed or reported.

Adoption Added as a Secondary Outcome

Two-month workplace use of ChatGPT was not an outcome in the October 22 plan and is added as a secondary outcome.

Inference Changed

The October 22 plan judged each hypothesis by whether a bootstrap interval excluded zero. The current plan uses Bayesian estimation and reports posterior summaries without any cutoff. The bootstrap is kept as a robustness check.

Posttask ChatGPT-Use Items Excluded as Covariates

The item asking whether the participant used ChatGPT on the second task, and its follow-up asked only of those who said yes, were both measured after the second task (Noy, 2023, ReadMe for Replication Pkg.pdf, pp. 4–5; Noy & Zhang, 2023b, pp. 2, 84). Conditioning on a

variable measured after assignment can bias the estimated effect (Montgomery et al., 2018), so neither is a covariate in any model, and use on the second task is reported by condition as a descriptive measure only. In their place, the adoption model includes prior use and awareness of ChatGPT, both asked before the first task and prespecified on October 25 before any outcome was analyzed.

Sensitivity Analyses Based on Posttask ChatGPT Use Removed

The October 22 plan listed a sensitivity analysis restricted by reported ChatGPT use on the second task and a sensitivity analysis that dropped participants who reported using ChatGPT on the first task. Both were removed on October 25 because they conditioned on items measured after assignment (Montgomery et al., 2018; Noy, 2023, ReadMe for Replication Pkg.pdf, pp. 4–5) and broke the intention-to-treat comparison.

Designation Sequence

The designation of Hypothesis 1 (writing quality) as the single primary estimand is withdrawn. The locked plan structure is restored unchanged: the five hypotheses form the planned family, Hypothesis 5 stays labeled secondary, no primary outcome is designated, and no multiplicity adjustment is made.

Exact Counts for Public Data

The locked plan’s minimum reportable cell size of 10 was withdrawn for this reanalysis of public Open Science Framework data (Noy & Zhang, 2023a). Under a plan amendment dated October 27, 2026, all counts, including exclusion flow, missingness, and per-condition n, are reported exactly, with no small-cell suppression, consistent with JARS-Quant guidance on reporting participant flow (Appelbaum et al., 2018).

Monte Carlo Error Criterion Narrowed

The earlier draft plan applied the Monte Carlo standard error criterion to every reported quantile. That criterion cannot be met for the tails of the credible interval at the planned number of draws, so the current plan applies it to the posterior median of each reported estimand and computes the Monte Carlo errors of the interval endpoints without using them as a criterion (Vehtari et al., 2021). The change was made before any model of the current plan was estimated.

Missing Baseline Covariates Handled by Complete Cases

The earlier draft plan imputed missing baseline covariates jointly within the Bayesian model. The current plan instead analyzes participants with all baseline covariates observed. The change was made before any model of the current plan was estimated.

Sampling Software Changed

On October 29, 2026, the models were moved from the CmdStan backend to the rstan backend of brms. The Stan models, priors, and random seeds are unchanged; the change is a software deviation.

Bootstrap Computed by a New Script

The bootstrap kept as a robustness check (see Inference Changed, above) is computed by a new script written for the current plan, not by rerunning the analysis code of the October 22 run.

Descriptive and Reliability Statistics Specified After the First Run

The baseline descriptive statistics, standardized differences, participant counts, and grader reliability statistics were specified after the October 22 run and before any model of the current plan was estimated.

Exploratory Analyses and Model Checks Not Built

On October 29, 2026, before any model of the current plan was estimated, the following items of the earlier plan were dropped: the main models in the exact-time condition, the grade models refit with the grader screen turned off, the models within each task-order stratum, and a median-absolute-deviation comparison for Hypothesis 3. These were not built under the frozen plan and are not reported, because building them now would add analyses specified after the October 22 result was known. Posterior and prior predictive checks and per-chain plots listed in the earlier plan were not built either; the convergence diagnostics and priors of every fit are reported instead.

Monte Carlo Error for the Multiple-Imputation Adoption Model

On October 29, 2026, before any model of the current plan was estimated on study data, a synthetic rehearsal showed that the Monte Carlo error of the multiple-imputation adoption estimate had been computed on the pooled chains of all 50 imputations, which counts the real differences between imputed data sets as sampling error. The criterion was kept (Monte Carlo standard error of the posterior median below 1% of the posterior standard deviation); for this model it is now computed within each imputed data set and combined across the 50 data sets, as was already done for R-hat and effective sample size.

Adoption Analyses Reestimated After a Code Correction

The adoption analyses produced no estimate in the planned real-data analysis on October 30, 2026: the complete-case Bayesian logistic model, fixed in the plan with four prior scenarios, and the adoption suite, which holds the multiple-imputation estimate (the primary adoption estimate) and the complete-case, tipping-point, and trimming comparisons. The frozen code averaged model predictions over all participants with an observed adoption response, and in the

tipping-point step over all participants, including participants with a missing baseline covariate (prior use or awareness of ChatGPT), for whom predictions are undefined. The complete-case logistic model stopped with an error in its summary step under all four prior scenarios. In the adoption suite, the multiple-imputation model converged and met the Monte Carlo error criterion, because the first imputed data set, over which its predictions are averaged, has no missing covariate; the complete-case model then converged but failed the Monte Carlo error criterion, whose value could not be computed, and its one permitted rerun with doubled iterations and a target acceptance rate of .999 gave the same result; the suite then stopped with an error before any result was written. The suite’s status record holds only the error message, so which of its two affected steps, the complete-case row or the tipping-point step, raised the error was not recorded; both were corrected. No posterior median or interval for adoption was saved or seen; the attempt records hold Monte Carlo diagnostics only, including the posterior standard deviation of the multiple-imputation risk difference. That record was written at 12:17 PT, before my authorization of the correction at 12:51 PT; the correction had been proposed earlier that day in a code-only review written before the record existed, and no record shows that the standard deviation was viewed before my authorization, although the records cannot rule out an unrecorded view. The plan defines complete-case analysis on all model variables, so this was an implementation error; it was not visible in the synthetic rehearsals, whose data had no missing covariates. With my authorization on October 30, 2026, and following guidance on correcting errors in planned analyses and reporting the correction next to the planned version (Lakens, 2024), the averaging set was corrected to the model’s complete-case sample in both the complete-case model and the tipping-point step, in a new postfreeze file first tested on synthetic data with injected missing covariates. The complete-case logistic model (all four prior scenarios)

and the adoption suite alone were then reestimated in a corrected follow-up analysis the same day, with unchanged models, priors, sampler settings, convergence and Monte Carlo error criteria, and failure rule. In brms 2.23.0, holding the initial compiled fit fixed still leaves per-imputation draws differing between runs with identical inputs, and the multiple-imputation estimate is reproducible within its Monte Carlo error, which the Monte Carlo error criterion bounds, whereas the complete-case estimates reproduce exactly. These adoption estimates were obtained after a postfreeze code correction and after the other results were available, and are reported as such. The corrected complete-case analysis used 276 participants, and the tipping-point analysis used 345 of the 359 participants in the primary sample (14 left out for a missing covariate).

Plan Version Ratified and Plan Version Run

On October 29, 2026, I ratified version 3 of the analysis plan and authorized a planned real-data analysis. Versions 3a, 3b, and 3c (dated October 29, 2026) and version 3d (dated October 30, 2026) were refinements made by AI agents working under my direction; version 3d was frozen before the planned real-data analysis, and on October 30, 2026, under a project rule recorded that morning from my instruction about automating the launch steps, the AI agents applied my October 29 ratification and authorization to it; I did not ratify version 3d separately. The AI agents launched the planned real-data analysis of October 30, 2026, under that authorization; only the corrected follow-up analysis (see Adoption Analyses Reestimated After a Code Correction, above) had an authorization specific to that read.

Earlier Modeling Attempts Withdrawn

On October 24, the first person-grouped split was discarded after its held-aside partitions were found stored in a location readable by other analyses in the project, from about 4:35 a.m.

Pacific time until the split was voided (about 10:42 a.m.). No read of those data was detected, according to the project's custody record. The same participants were rerandomized, and models were built on the new training partition through several protocol revisions. On October 24 at 12:49 p.m. Pacific time, I withdrew all models and outputs from those revisions. Neither split is used here: the models reported in Results were estimated in two planned analyses on October 30, 2026, on all 359 participants in the primary sample, drawn from the 453 published participants, so the sample contains the participants whose outcomes were analyzed on October 22 and in the withdrawn revisions.

Exposure to Earlier Results

While the current plan was drafted, three items were seen: a one-line summary of an earlier frequentist result for the October 22 hypotheses; cross-validated predictive summaries from a withdrawn revision; and a pooled count (not split by condition) of the ChatGPT-use item from the two-week follow-up, an earlier survey wave than the two-month follow-up analyzed here. That pooled count was seen while the timing of the ChatGPT-use item was being checked. No row-level outcome data and no output files from the two withdrawn revisions were opened in drafting the plan. Earlier analyses on overlapping data also modeled outcomes in a training partition (see Limitations).

Table 1

Baseline Characteristics by Condition

Characteristic	ChatGPT access	Comparison	Std. diff.
Occupation, *n* (%)			
Marketing	20 (9.2)	27 (11.5)	−0.08
Grant writing	37 (17.0)	37 (15.7)	0.03
Consulting	25 (11.5)	30 (12.8)	−0.04
Data analysis	23 (10.6)	26 (11.1)	−0.02
Human resources	23 (10.6)	14 (6.0)	0.17
Management	90 (41.3)	101 (43.0)	−0.03
Employment status: full-time, *n* (%)	182 (86.7)	176 (77.9)	0.23
Tenure in occupation, *M* (*SD*)	10.07 (8.45)	10.49 (9.24)	−0.05
Weekly hours, *M* (*SD*)	38.29 (9.62)	37.74 (10.73)	0.05
Did similar task in job, *n* (%)	144 (67.0)	163 (69.7)	−0.06
Prior use of ChatGPT before Task 1[a], *n* (%)	69 (32.9)	69 (30.4)	0.05
Aware of ChatGPT before Task 1[a], *n* (%)	146 (68.2)	168 (71.8)	−0.08
Reported ChatGPT use, Task 2[b,c], *n* (%)	173 (80.5)	21 (9.0)	—
Task realism, *M* (*SD*)	4.01 (0.76)	4.07 (0.86)	−0.07
Payment: linear, *n* (%)	81 (37.2)	90 (38.3)	—
Payment: convex, *n* (%)	91 (41.7)	97 (41.3)	—
Payment: exact-time, *n* (%)	46 (21.1)	48 (20.4)	—
Task order: Task A first, *n* (%)	107 (49.1)	132 (56.2)	−0.14
First-essay writing grade (1–7), *M* (*SD*)	4.05 (1.48)	3.88 (1.51)	0.11
First-essay content grade (1–7), *M* (*SD*)	3.77 (1.49)	3.64 (1.53)	0.09
First-essay originality grade (1–7), *M* (*SD*)	3.74 (1.39)	3.68 (1.42)	0.04
First-essay overall grade (1–7), *M* (*SD*)	3.77 (1.46)	3.63 (1.46)	0.10
Felt effectiveness, Task 1 (1–10), *M* (*SD*)	6.90 (2.13)	6.89 (2.09)	0.00
Enjoyment, Task 1 (1–10), *M* (*SD*)	6.34 (2.37)	6.30 (2.54)	0.02
Two-month follow-up completed[b], *n* (%)	174 (79.8)	186 (79.1)	—

Note. $n = 218$ (ChatGPT access) and 235 (comparison). Percentages, means, and standard deviations are computed among the participants who answered each item, so for items with missing responses the denominator is smaller than n. Age, gender, and race or ethnicity were not released in the archival data. Standardized differences are shown for characteristics measured before assignment with two levels or a mean; the payment condition has three levels and the two rows marked b were measured after assignment, so those cells are left blank. Grader agreement on writing grades: ICC(1) = .38, ICC(k) = .63; mean graders per essay = 2.72. ICC = intraclass correlation coefficient; Std. diff. = standardized difference.

[a] Asked before Task 1; adoption-model covariate only (missing values imputed). [b] Measured after assignment; descriptive only. [c] Options: *yes*, *no*, *does not know ChatGPT*.

Table 2

Samples and Outcome Availability by Condition

	ChatGPT access	Comparison	Total
Randomized	218	235	453
Primary sample (linear and convex conditions)	172	187	359
Exact-time condition (exploratory)	46	48	94
Writing, content, and originality grades available	167	186	353
Felt effectiveness and enjoyment available	170	187	357
Two-month adoption item answered	139	150	289

Note. Counts correspond to Figure 1. Small counts are reported exactly.

Table 3

Posterior Summaries for Hypotheses 1–5

Hypothesis and estimand	*Mdn* [95% CrI]	*P*(direction)
H1: condition × first-task writing grade	−0.159 [−0.317, −0.002]	.976
H2: content minus writing interaction	−0.01 [−0.14, 0.11]	.423
H3: originality minus writing log *SD* ratio	−0.01 [−0.14, 0.12]	.576
H4: condition, felt effectiveness	0.12 [−0.06, 0.29]	.905
H4: condition × first-essay overall grade	−0.11 [−0.27, 0.06]	.894
H5: condition, enjoyment	0.44 [0.29, 0.60]	1.000
H5: condition × first-essay overall grade	0.04 [−0.11, 0.19]	.299

Note. Primary sample. First-task writing, content, and overall grades are centered at the primary-sample mean and divided by the comparison group's standard deviation of the same grade, second-task writing and content grades are rescaled with the mean and standard deviation of the matching first-task grade, and second-task originality grades and the felt effectiveness and enjoyment ratings of both tasks are standardized to a mean of 0 and a standard deviation of 1 in the primary sample. Hypothesis 5 is secondary. Estimates are on the standardized scale defined in Method, so no separate standardized column is shown; the Hypothesis 3 estimate is a difference of log SD ratios. Analysis ns (complete case): Hypothesis 1 = 352; Hypothesis 2 = 352; Hypothesis 3 = 353; Hypothesis 4 = 352; Hypothesis 5 = 352. Mdn = posterior median; CrI = 95% equal-tailed credible interval; *P*(direction) = posterior probability that the effect is in the predicted direction; *SD* = standard deviation.

Table 4

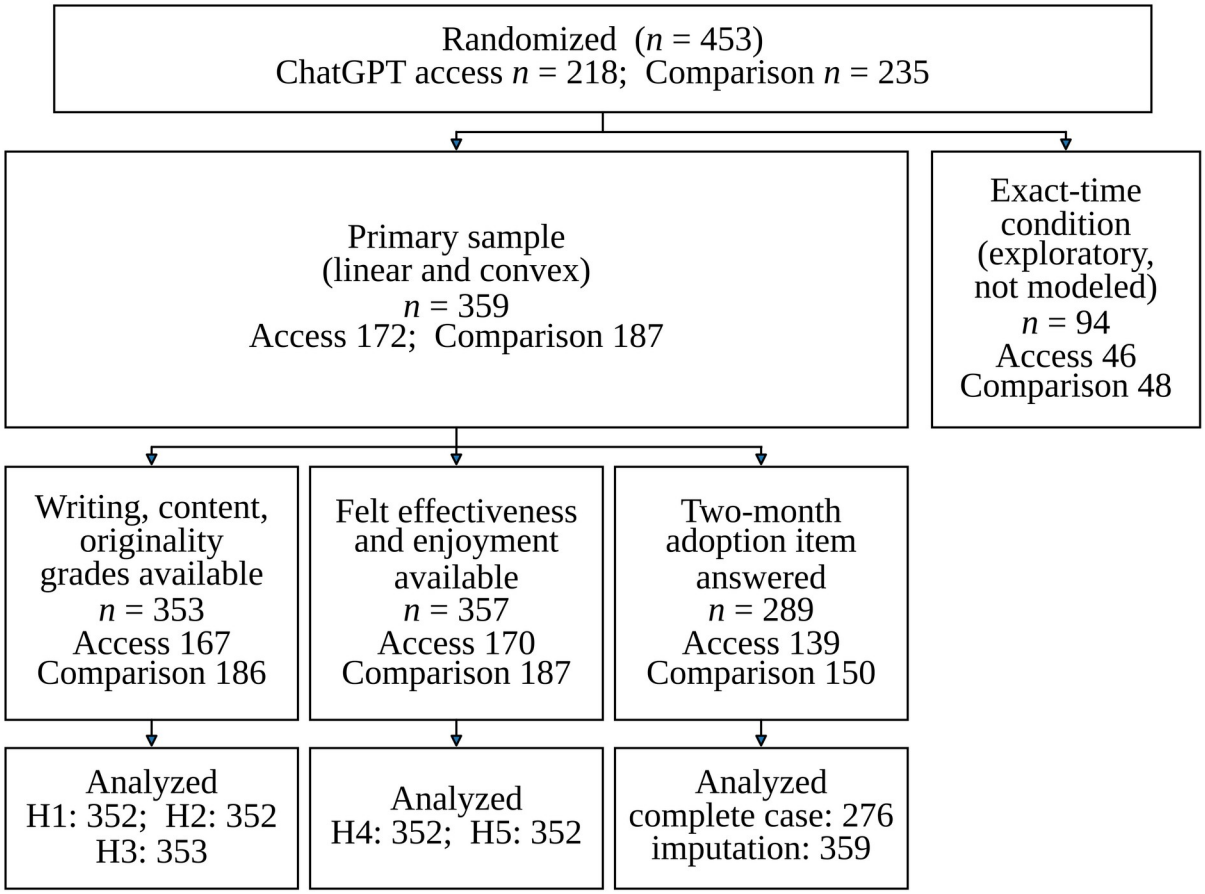
Two-Month Adoption by Condition

Analysis	Access %	Comparison %	RD, pp [95% CrI]	*P(RD > 0)*
MI (primary)	42.2	28.8	13.4 [3.0, 23.5]	.994
Complete case	41.8	28.3	13.5 [3.1, 23.7]	.995
Tipping-point range	—	—	10.1 to 17.2	—
Lee (2009) trim	—	—	14.3 to 15.7	—

Note. Secondary outcome. Condition percentages are model-standardized margins; their difference need not equal the median risk difference. Tipping-point and trimming rows give ranges of the risk difference; condition percentages and P are not defined for them, so those cells are left blank. The multiple-imputation (primary) estimate uses $n = 359$ and the complete-case estimate $n = 276$. CrI = 95% equal-tailed credible interval; MI = multiple imputation; P = posterior probability that the risk difference is greater than zero; pp = percentage points; RD = risk difference.

Figure 1

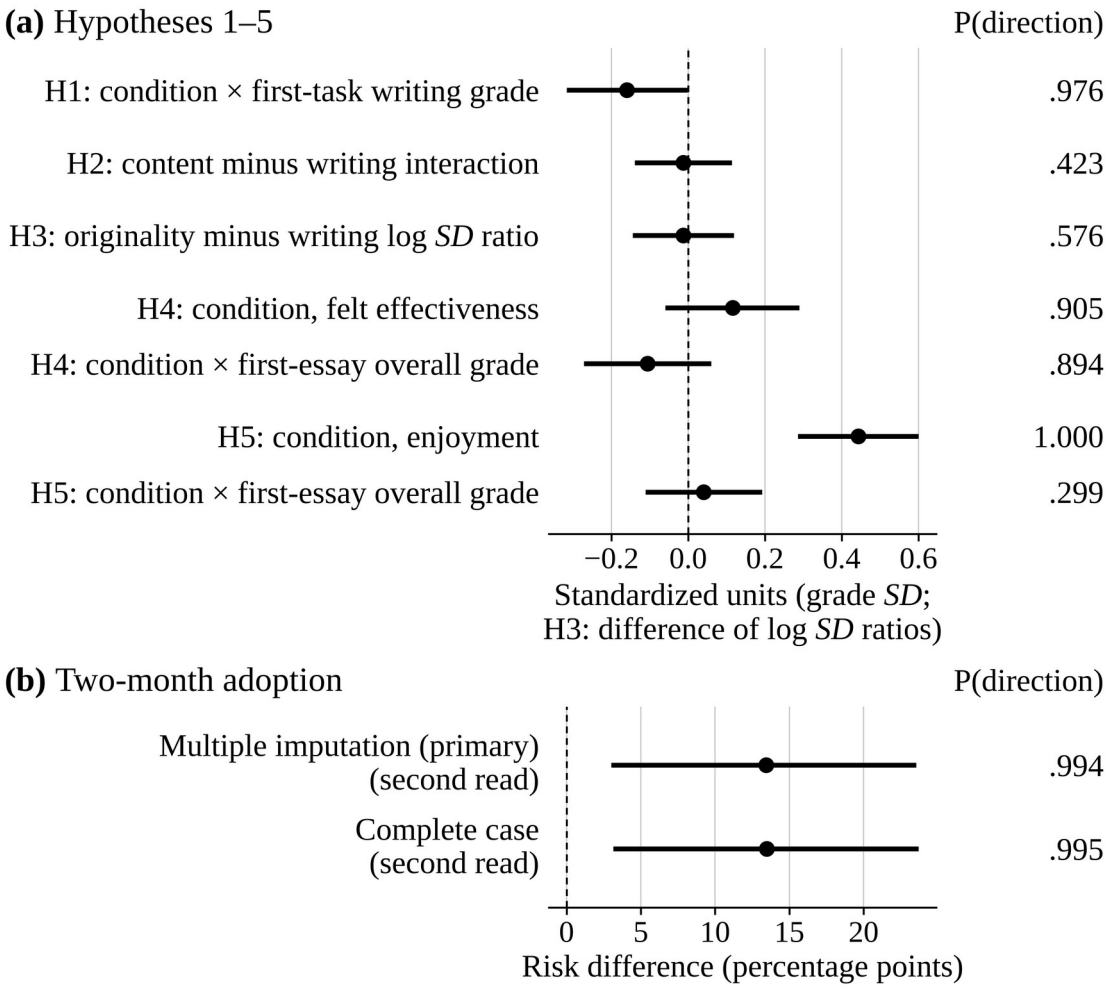
Participant Flow (CONSORT 2025)



Note. The flow branches at randomization. Of the 453 randomized participants (218 with ChatGPT access and 235 in the comparison condition), 359 were in the primary sample, the linear and convex incentive conditions (172 access, 187 comparison), and 94 were in the exact-time condition (46 access, 48 comparison), a separate branch that is described but not modeled. From the primary sample, three parallel boxes, not nested in one another, show the participants with each outcome available: writing, content, and originality grades for 353 (167 access, 186 comparison), felt effectiveness and enjoyment for 357 (170 access, 187 comparison), and the two-month adoption item, answered by 289 (139 access, 150 comparison). Below each availability box, an analyzed box gives the participants in each model, those with the outcome and all baseline covariates observed (see Participants): the models for Hypotheses 1 and 2, Hypothesis 3, and Hypotheses 4 and 5, and for adoption the complete-case model (276) and the multiple-imputation model, which uses all 359 participants in the primary sample.

Figure 2

Effects of ChatGPT Access on Grades, Ratings, and Adoption



Note. Primary sample (n = 359; the analysis ns for each model are given in Tables 3 and 4 and Figure 1). Values come only from fits that met every convergence and Monte Carlo error criterion in Method; estimates for Hypotheses 1, 2, 4, and 5 are in the standardized units defined in Method, the Hypothesis 3 estimate is a difference of log standard-deviation ratios, and adoption is a risk difference in percentage points from the corrected follow-up analysis (see Appendix, Adoption Analyses Reestimated After a Code Correction). Circles mark posterior medians, horizontal lines mark 95% equal-tailed credible intervals, the dashed vertical line marks zero, and the column at right gives *P*(direction); an estimand without an estimate is labeled not estimable. *P*(direction) = posterior probability that the effect is in the predicted direction; CrI = 95% equal-tailed credible interval; *SD* = standard deviation.